

INDIAN INSTITUTE OF TECHNOLOGY KANPUR

Department of Management Sciences

RESEARCH PROPOSAL

Beyond Population: Understanding the Socioeconomic Drivers of UPI Adoption in India

Proposed Research Work under the Supervision of Prof. Suwendu Naskar

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1. Abstract

India's Unified Payments Interface (UPI) has become the dominant mode of digital retail payment in the country, with transaction volumes rising sharply since its launch. However, adoption across Indian states remains uneven, and most existing analyses attribute this variation largely to differences in population size. This project proposes a more comprehensive investigation of the socioeconomic and infrastructural determinants of state-wise UPI adoption, including income levels, literacy, and telecom infrastructure. Using a panel dataset constructed from **NPCI, RBI, Census, PLFS, and telecom** sources, the study will apply both traditional statistical methods (correlation and regression analysis) and machine learning models (Random Forest, XGBoost) to identify and rank the drivers of adoption. Model interpretability will be established through SHAP analysis, and K-Means clustering will be used to group states by digital payment behaviour. The project will culminate in a composite Digital Payment Readiness Index (DPRI) that can be used to benchmark states and inform targeted policy interventions for improving digital financial inclusion.

2. Introduction

Since its launch by the National Payments Corporation of India (NPCI) in 2016, UPI has fundamentally altered the landscape of retail digital payments in India, enabling real-time, low-cost transactions between bank accounts through a single mobile interface. Monthly UPI transaction volumes have grown from a few million to several billion, and UPI is now central to India's digital financial inclusion agenda. Despite this aggregate success, the pace and depth of adoption differ considerably across states.

A large part of the existing literature on UPI has examined individual-level adoption using technology-acceptance frameworks such as the Diffusion of Innovation theory and UTAUT, focusing on user perceptions of convenience, trust, and ease of use. Comparatively fewer studies have examined why adoption varies systematically across geographies at a macro level, and where such studies exist, they tend to treat population size as the principal explanatory variable. This project addresses that gap by examining a broader set of state-level socioeconomic and infrastructural factors — income, literacy, and telecom penetration — as determinants of UPI adoption, and by applying machine learning methods that can capture non-linear relationships that traditional regression may miss.

3. Problem Statement

Current understanding of state-wise UPI adoption disparities is limited by an over-reliance on population as an explanatory factor, which obscures the role of income levels, literacy, and digital infrastructure in shaping adoption patterns. Without a clearer, quantitatively grounded understanding of these drivers, policy efforts to close the digital payment gap between states risk being poorly targeted. This project addresses the question: **which socioeconomic and infrastructural factors best explain the variation in UPI adoption across Indian states, and how can these factors be combined into a single, interpretable readiness measure?**

4. Research Objectives

- To study state-wise variation in UPI adoption using publicly available transaction data.
- To identify and quantify socioeconomic determinants of adoption, including income, literacy, and telecom infrastructure.
- To build predictive machine learning models (Random Forest, XGBoost) for state-level UPI transaction volumes.
- To compare the performance and interpretability of traditional statistical methods against ML models.
- To develop a composite Digital Payment Readiness Index (DPRI) for Indian states using clustering and index-construction techniques.

5. Scope of the Study

The study is confined to state- and union-territory-level analysis within India, using publicly available secondary data covering recent years for which NPCI, RBI, Census/PLFS, and telecom datasets are available. The analysis focuses on structural and socioeconomic determinants of adoption rather than individual-level behavioural or psychological factors, which have already been extensively studied in the existing literature. District-level or transaction-category-level analysis (e.g., person-to-person versus person-to-merchant) is treated as a possible extension rather than a core objective, subject to data availability and time constraints.

6. Literature Background

Existing research on UPI adoption can broadly be grouped into three strands. The first strand applies technology-adoption frameworks at the individual level: studies using the Diffusion of Innovation theory find that relative advantage, complexity, and observability significantly shape users' intention to adopt UPI, while UTAUT-based studies show that demographic factors such as age, education, and income influence adoption and usage intensity. The second strand examines UPI's macroeconomic and financial-inclusion implications, including its contribution to formalisation of economic activity, reduction in payment costs, and its role alongside broader digital-inclusion initiatives; work using NPCI-based state-level econometric specifications has linked UPI growth to improvements in payment infrastructure supply. The third strand looks specifically at inequality and rural-urban divides in adoption, with recent district-level studies finding that income inequality has a pronounced effect on digital payment penetration, and rural case studies identifying income, education, and gender as significant predictors of usage. This project builds on the second and third strands by moving from individual or district-level analysis to a state-level panel framework that explicitly incorporates literacy and telecom infrastructure alongside income, and by using machine learning methods to capture non-linear interactions between these determinants that regression-based studies may not fully reveal.

7. Data Sources

The study will draw on multiple publicly available secondary data sources, integrated into a single state-year panel dataset:

Data Source	Variables of Interest
NPCI Monthly UPI Statistics	Transaction volume and value at national and state level
RBI Bulletins and Reports	Macroeconomic indicators, banking penetration, digital payment trends
Census of India (2011) / Population Projections	State-wise population, urbanization, and literacy baselines
Periodic Labour Force Survey (PLFS), MoSPI	State-wise literacy, employment, and income-related indicators
Department of Telecommunications (DoT)	Tele-density, internet subscriber base, and smartphone penetration by state
State-wise Per Capita Income (MoSPI / CSO)	Net State Domestic Product and per capita income series

8. Research Methodology

8.1 Literature Review

A structured review of academic and industry literature on UPI adoption, digital financial inclusion, and index construction methods (e.g., composite indices used by RBI and NITI Aayog) to ground the study's variable selection and modelling choices.

8.2 Data Collection and Preprocessing

Collection of state-wise time series data from NPCI, RBI, Census/PLFS, and DoT sources; harmonization of state codes and naming conventions; handling of missing values and outliers; and construction of a merged state-year panel dataset.

8.3 Feature Engineering and Exploratory Data Analysis

Derivation of per-capita and normalized indicators (e.g., UPI transactions per capita, tele-density ratios), followed by exploratory analysis to examine distributions, trends over time, and state-wise disparities.

8.4 Statistical Analysis

Correlation analysis to identify pairwise relationships between socioeconomic variables and UPI adoption, followed by multiple linear regression to establish a baseline model and test the statistical significance of individual determinants.

8.5 Machine Learning Modelling

Development of Random Forest and XGBoost regression models to predict state-level UPI adoption, with hyperparameter tuning via cross-validation. Model performance will be compared against the regression baseline using R^2 , RMSE, and MAE.

8.6 Interpretability Analysis

Feature importance rankings from the tree-based models, supplemented by SHAP (SHapley Additive exPlanations) values, to interpret the direction and magnitude of each determinant's contribution to predicted adoption.

8.7 Clustering and Index Construction

K-Means clustering of states based on standardized socioeconomic and adoption features to identify natural groupings, followed by construction of a composite Digital Payment Readiness Index (DPRI) that combines the underlying indicators into a single, interpretable score for cross-state comparison.

8.8 Visualization and Interpretation

Consolidation of findings through choropleth maps, bar charts, and cluster plots, accompanied by a written interpretation connecting statistical results to policy-relevant insights.

9. Expected Outcomes

- A clearer, evidence-based understanding of which socioeconomic and infrastructural factors most strongly influence UPI adoption across Indian states.
- A comparative assessment of the predictive accuracy and interpretability of traditional regression versus ensemble ML methods for this problem.
- Identification of state clusters with similar digital payment adoption profiles, useful for targeted policy design.
- A reusable Digital Payment Readiness Index that can be updated as new data becomes available.

10. Deliverables

- A clean, integrated state-year panel dataset combining all data sources used in the study.
- Trained and validated predictive models (regression, Random Forest, XGBoost) with documented performance metrics.
- A comparative performance analysis of statistical and machine learning approaches.

- The constructed Digital Payment Readiness Index (DPRI) with state-wise scores and rankings.
- A written research report summarizing methodology, findings, and policy implications.
- A final presentation summarizing the project for the supervisor.

11. Timeline

The project is planned over a 12-week period, structured as follows:

Week	Key Activities	Milestone / Output
1-4	Literature review; finalization of research questions; review of prior UPI adoption studies	Literature review note finalized
5-6	Data collection: NPCI UPI statistics, RBI indicators, Census/PLFS, telecom data, state income data	Raw dataset compiled
7-8	Data cleaning, state-code harmonization, handling missing values, merging multi-source panel	Clean master dataset (state-year panel)
9	Feature engineering (per-capita transactions, infrastructure ratios); exploratory data analysis	EDA report with visualizations
10	Correlation analysis; baseline multiple linear regression on socioeconomic determinants	Baseline regression results
11	Random Forest model development and hyperparameter tuning	Trained Random Forest model
12	XGBoost model development; comparison of ML models with regression baseline	Model comparison table (R^2 , RMSE, MAE)
13	Feature importance analysis and SHAP interpretation of best-performing model	SHAP summary plots and interpretation
14	K-Means clustering of states; construction of Digital Payment Readiness Index (DPRI)	DPRI scores and state clusters
15-16	Consolidation of visualizations; robustness checks; draft research report	First draft of research report
17-18	Report refinement, supervisor feedback incorporation, preparation of final presentation	Revised report and slide deck

19-20	Final review, submission of report, and presentation to supervisor	Final report and presentation delivered
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12. Software and Tools

- Python (pandas, NumPy, scikit-learn, XGBoost, SHAP) for data processing, modelling, and interpretability analysis.
- Jupyter Notebook for iterative analysis and documentation of the modelling workflow.
- Matplotlib / Seaborn / Plotly for exploratory and result visualization.
- Microsoft Excel for preliminary data checks and cross-verification of aggregated figures.
- Git/GitHub for version control and reproducibility of the analysis pipeline.

13. Expected Contributions

This project is expected to contribute a state-level, multi-source empirical perspective on UPI adoption that goes beyond population as an explanatory variable, and to demonstrate how machine learning methods can complement, rather than replace, traditional statistical analysis in a policy-relevant economic context. The proposed Digital Payment Readiness Index is intended as a practical, extensible tool that could be adapted by researchers or policymakers to track digital payment readiness over time, and the project's methodology and findings are also intended to inform the researchers' ongoing work and future academic or applied engagements in the fintech and data analytics domain.

14. References

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